TECHNICAL DILIGENCE REVIEW - RECORD 002

Helioscope Robotics, Inc. - diligence file reference TECHNICAL_DILIGENCE-002

1. Terms of reference, method and interview programme

As instructed, Calder Rowe Engineering Advisory confined report TDR-2025-09 to the reliability record, incident history and post-incident follow-through of the fleet platform; three adjacent questions raised during the work are listed as follow-ups rather than answered here. This report was prepared by Calder Rowe Engineering Advisory for Ardent Peak and covers the reliability record, incident history and post-incident follow-through of the fleet platform at Helioscope Robotics, Inc. as at 2025-09-05. The present report concludes that the safety work is the strongest part of the engineering record and the delivery process is the weakest. S. Alderton would summarise the report at TDR-2025-09 in one sentence: the safety work is the strongest part of the engineering record and the delivery process is the weakest. The reviewer's tone in the report at TDR-2025-09 is deliberately flat; a finding is not made more serious by the words around it.

Work on the report at TDR-2025-09 ran over 3 weeks and included 19 interviews and read-only access to the engineering repositories. In this report the reviewer distinguishes what was verified from what was represented, and marks each conclusion accordingly. Nothing in report TDR-2025-09 supports a retention or an escrow; the findings are operational and they are funded. Calder Rowe Engineering Advisory notes that 3 of the matters in the present report are capable of being closed before completion if the company is asked in time. The company's engineering leadership saw every finding in TDR-2025-09 before delivery and disputed two of them.

Minutes of every the present report interview were returned to the participant, and three came back with substantive corrections. The reviewer found the interview evidence for the report at TDR-2025-09 candid, and in two cases more candid than the written record. The principal limitation on TDR-2025-09 is time: 3 weeks was enough to test the control path and not enough to test the reporting path. The reviewer notes that the report at TDR-2025-09 was prepared while three remediation programmes were in flight, so several positions will have moved by completion. The reader of this report should treat each conclusion as bounded by the section it appears in.

Nothing in the present report is presented as verified on the strength of an interview alone. S. Alderton has left several concerns in this report at limited evidence strength rather than dropping them or dressing them up. For this report the reviewer sampled 6 sites, including the largest deployment and the newest. S. Alderton chose the sample for TDR-2025-09 after reading the dependency map, so the selection is not proportionate to headcount or to code volume. S. Alderton reports management's own position wherever it differs from the reviewer's, and does not merge the two.

One limitation of TDR-2025-09 should be stated plainly: the reviewer had no production access and relied on the company's own measurements for live latency. S. Alderton did not visit the two most recently commissioned sites, so the commissioning quality there is taken from the company's records alone. Calder Rowe Engineering Advisory asked the same four questions of every team interviewed for the report at TDR-2025-09, which is how the inconsistencies below were found. The interview programme behind the report at TDR-2025-09 covered 6 teams and produced 15 follow-up requests, of which most were satisfied. S. Alderton has resisted rounding the findings towards a comfortable conclusion in either direction.

The population for this report was the service inventory as at 2025-09-05; 19 entries were examined and the rest were reviewed only at inventory level. Sampling in the report at TDR-2025-09 covered 19 of the 34 catalogued services and 6 of the 18 live sites. The convention throughout report TDR-2025-09 is that an absence of evidence is reported as such rather than as evidence of absence. Grading in TDR-2025-09 is deliberately coarse, because a finer scale would imply a precision the underlying evidence does not carry. S. Alderton has separated what the company has built from what the company can demonstrate about what it has built.

Schedule I - Interview programme and follow-up requests arising (TDR-2025-09)

Date	Participant	Role	Principal topic	Follow-ups raised	Follow-ups open
2025-09-25	Marcus Duplessis	head of engineering	Analytics Warehouse Loader	3	0
2026-04-18	Marcus Duplessis	engineering owner	E-Stop Chain	5	0
2025-08-26	Owen Brantley	engineering owner	Motion Planner	1	0
2025-09-08	Nadia Semenova	team lead	Costmap Fusion	5	2
2026-04-28	Farida Cisse	security engineer	WMS Connector	1	1

Date	Participant	Role	Principal topic	Follow-ups raised	Follow-ups open
2025-11-15	Wanda Njoku	head of engineering	Simulation Harness	5	2
2026-08-11	Yuki Harada	site reliability lead	Safety Controller	4	1
2025-09-14	Peter Halvorsen	engineering owner	Fleet Health Monitor	5	3
2026-06-26	Nathan Oduya	site reliability lead	Map Authoring Studio	4	4
2025-10-29	Nadia Semenova	site reliability lead	Costmap Fusion	4	3
2025-11-08	Wanda Njoku	engineering owner	Localization Engine	1	0
2026-03-10	Grace Lindqvist	safety engineer	Hardware-in-the-Loop Rig	2	1
2026-06-21	Anjali Sekhon	release manager	Event Bus	5	0
2026-08-10	Marta Kellerman	safety engineer	Costmap Fusion	5	4
2026-09-28	Wanda Njoku	release manager	Alarm Router	2	0
2026-04-01	Sam Oyelaran	safety engineer	OTA Updater	2	2
2025-10-03	Theo Bassett	quality lead	Hardware-in-the-Loop Rig	1	0
2026-09-17	Priya Raghunathan	head of engineering	Wheel Odometry Filter	5	3
2026-07-23	Nadia Semenova	site reliability lead	Map Authoring Studio	2	2
2026-04-24	Marcus Duplessis	safety engineer	Fleet Health Monitor	4	4
2026-09-19	Anjali Sekhon	team lead	Telemetry Ingest	1	1
2026-09-23	Colin Achebe	release manager	Speed and Separation Monitor	3	1
2026-05-10	Marcus Duplessis	site reliability lead	Fleet Allocator	4	3
2026-01-24	Theo Bassett	team lead	Localization Engine	2	1
2026-01-26	Peter Halvorsen	principal engineer	Motion Planner	3	1
2025-09-14	Farida Cisse	site reliability lead	Traffic Manager	1	1
2025-10-03	Theo Bassett	site reliability lead	Alarm Router	1	1
2026-08-29	Yuki Harada	principal engineer	Telemetry Ingest	4	1
2025-10-24	Daniel Okafor	principal engineer	Speed and Separation Monitor	3	0
2025-12-18	Peter Halvorsen	engineering owner	Gripper Controller	3	2

2. INC-2025-090 and the incident population examined

Mitigation of hardware fault event INC-2025-090 took immediate stop, 22 m to return to service and full resolution took 41 days. For hardware fault incident INC-2025-090 the reviewer holds limited; the log retention window had closed before the reviewer asked for the raw data evidence and reports the finding at that strength. Underneath Safety Controller incident INC-2025-090 lies the design decision that a marginal connector on one channel produced intermittent readings inside the accepted noise band. The immediate event in hardware fault incident INC-2025-090 was that a protective stop was triggered by a false positive on one safety channel while the second channel read clear, dated 2025-07-19. The corrective action for INC-2025-090 is to add the connector torque check to incoming inspection and retrofit the affected harness batch, and 2025-09-30 is the date the team gave. INC-2025-090 is reported at the evidence strength available and has not been promoted to fit the narrative.

Verification of the incident INC-2025-090 remediation gives the inspection change landed on 2025-08-04; nine units await retrofit. Fathom Mobility incident INC-2025-090 affected one site and 3 robots. Safety Controller owns Safety Controller incident INC-2025-090, which the reviewer classes as hardware fault. Drawing the evidence in Fathom Mobility incident INC-2025-090 together, the incident is unremarkable in itself and revealing about the observability of the surrounding system. Attention came to Fathom Mobility incident INC-2025-090 when the robot self-reported the fault and the safety case review picked it up. hardware fault event INC-2025-090 is included because a reader of TDR-2025-09 should be able to disagree with the inference without repeating the work.

Video Snippet Service accounts for INC-2025-152, a data retention event. Data retention event INC-2025-152 traces back to a single fact: the lifecycle rule was applied per storage bucket and one region's bucket was created later without it. Reading Video Snippet Service incident INC-2025-152 from its first entry, on 2025-12-04 one region retained operator video clips for ninety days against a thirty-day commitment. The commitment register shows internal incident INC-2025-152 against an undertaking to apply the lifecycle rule through the configuration baseline and delete the over-retained clips, with Anjali Sekhon named and 2026-01-31 committed. On incident INC-2025-152 the evidence is moderate; the narrative is consistent but the reviewer could not independently reproduce it; the severity assessment is

stated separately below. data retention incident INC-2025-152 was reviewed against the position twelve months earlier, and the direction of travel is favourable.

The contributing factor recorded against incident INC-2025-152 is that bucket creation was not covered by the configuration baseline check. The reviewer's reading of internal incident INC-2025-152 is that an operational workaround masked the defect for longer than the incident record suggests. Incident INC-2025-152 touched no production site and no robot directly, the exposure being to internal systems, and internal was the affected account. INC-2025-152 closure status is the rule landed on 2026-01-10; deletion of the historical clips is open. Data retention event INC-2025-152 was detected because an internal privacy review found the discrepancy. INC-2025-152 was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

Reported detection applies to data integrity event INC-2025-058, since the customer reported missing picks the same afternoon. Incident INC-2025-058 took 2 h 35 m to stabilise and 21 days to close out. On 2025-05-09 the record for incident INC-2025-058 opens: two tenants reused the same order identifier namespace and duplicate suppression discarded live orders. Root cause in WMS Connector incident INC-2025-058 was that duplicate suppression keyed on a customer-supplied identifier alone rather than on tenant and identifier. Drawing the evidence in Dunbar Industrial incident INC-2025-058 together, the same class of defect can recur in two sibling components that were not changed. data integrity event INC-2025-058 is the clearest instance in TDR-2025-09 of engineering practice being better than the documentation suggests.

Dunbar Industrial felt WMS Connector incident INC-2025-058, which affected 2 site or tenant with no robot taken out of service. The contributing factor recorded against data integrity event INC-2025-058 is that the integration specification allowed the identifier to be tenant-scoped but did not require it. Data integrity event INC-2025-058 belongs to the data integrity group and sits with WMS Connector. Grading data integrity event INC-2025-058 on evidence rather than on consequence gives moderate, resting on the team's own post-mortem and one supporting log extract. Data integrity event INC-2025-058 is recorded as follows on delivery: landed on 2025-05-30; the backfill was completed on 2025-06-03. incident INC-2025-058 is the reason one section of TDR-2025-09 concludes narrowly rather than generally.

Charge Scheduler incident INC-2025-052 classifies as a scheduling event against Charge Scheduler. Evergreen Warehousing felt INC-2025-052, which affected one site and 44 robots. On delivery, incident INC-2025-052 shows this: landed on 2025-06-11 with a nightly reconciliation report. On 2025-04-21 the record for Evergreen Warehousing incident INC-2025-052 opens: an early shift end left opportunity-charge windows reserved and the following shift started with a depleted fleet. Underneath scheduling incident INC-2025-052 lies the design decision that the scheduler treated a shift calendar entry as immutable once the window had opened. Charge Scheduler incident INC-2025-052 is not material on its own and belongs in this report because of what it reveals about the surrounding process.

INC-2025-052 would have been shorter or smaller had it not been that the site had changed its shift pattern without notifying the deployment team. The corrective action for INC-2025-052 is to release charge windows on a shift-end event and reconcile the calendar nightly, and 2025-06-30 is the date the team gave. Evidence strength on the scheduling incident INC-2025-052 analysis is limited, being a single account corroborated only by the ticket timeline. Reported detection applies to INC-2025-052, since the site's own shift report, escalated the next morning. Mitigation of scheduling event INC-2025-052 took 4 h 10 m and full resolution took 16 days. Evergreen Warehousing incident INC-2025-052 rests on a single source, and the reviewer has said so rather than implying corroboration.

Grading incident INC-2025-110 on evidence rather than on consequence gives strong on the cause and limited on the contributing factor. Data integrity incident INC-2025-110 classifies as a data integrity event against Analytics Warehouse Loader. INC-2025-110 closure status is landed on 2025-09-21 with a reprocessing run for the affected week. What data integrity incident INC-2025-110 tells the reviewer is that the fix is real and the regression protection is not, so recurrence remains possible. A contributing factor in data integrity incident INC-2025-110 was that the partition completeness check was written but never wired into the load gate. INC-2025-110 illustrates why TDR-2025-09 separates the severity of a finding from the confidence behind it.

Root cause in Analytics Warehouse Loader incident INC-2025-110 was that the loader treated a missing partition as empty rather than as incomplete. Detection came through the customer challenged the report three days later, which places data integrity incident INC-2025-110 on the reported side of the ledger. The blast radius of Kestrel Medical Supply incident INC-2025-110 was 1 site or tenant with no robot taken out of service. The commitment register shows

INC-2025-110 against an undertaking to fail the load on an incomplete partition and reprocess the affected days, with Farida Cisse named and 2025-10-31 committed. Time to mitigate data integrity event INC-2025-110 was not applicable; reporting only, and time to resolve was 18 days. Analytics Warehouse Loader incident INC-2025-110 is treated as material to the investment case rather than as a routine engineering item.

After availability incident INC-2025-071, Erik Palmgren took the action to add a leadership and under-replicated partition check to the broker upgrade runbook by 2025-07-31. Availability event INC-2025-071 is recorded as follows on delivery: landed on 2025-07-18 and exercised in the September failover drill. So far as reach is concerned, Telemetry Ingest incident INC-2025-071 involved 6 sites and 214 robots. Availability incident INC-2025-071 traces back to a single fact: a broker partition lost its leader during a rolling upgrade of the streaming cluster. Reading Telemetry Ingest incident INC-2025-071 from its first entry, on 2025-06-11 a regional ingest endpoint failed over and edge buffers filled during a fifty-two minute outage. incident INC-2025-071 is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

In incident INC-2025-071, the upgrade runbook did not require a partition leadership check between broker restarts, which is a process finding rather than a code finding. Telemetry Ingest incident INC-2025-071 classifies as a availability event against Telemetry Ingest. Alerted detection applies to INC-2025-071, since the ingest availability alert fired within ninety seconds. Telemetry Ingest incident INC-2025-071 supports one conclusion and not another: an operational workaround masked the defect for longer than the incident record suggests. The reviewer grades the incident INC-2025-071 evidence as moderate, resting on the team's own post-mortem and one supporting log extract and has not upgraded the finding to compensate. availability incident INC-2025-071 carries a follow-up request that was satisfied after the draft of TDR-2025-09 was circulated.

Multiple sites incident INC-2025-124 shows a short mitigation and a long tail: 1 h 08 m against 29 days. Alerted detection applies to incident INC-2025-124, since the console latency alert fired, though the underlying cause took a day to attribute. The blast radius of consistency incident INC-2025-124 was 5 sites and 178 robots. INC-2025-124 was raised on 2025-10-01 after a shard rebalance served shadow state up to ninety seconds stale without marking it stale. The reviewer checked the consistency incident INC-2025-124 commitment against the code and release records; landed on 2025-10-30; the load test was added on 2025-11-12. consistency incident INC-2025-124 was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

Robot Shadow Store incident INC-2025-124 is not read as an isolated failure: management's severity rating is defensible and its remediation priority is not. Contributing to consistency incident INC-2025-124, and not repaired by the remediation, is that no test exercised a shard rebalance while the cluster was under load. Evidence strength on the consistency event INC-2025-124 analysis is moderate, resting on the team's own post-mortem and one supporting log extract. Robot Shadow Store owns consistency incident INC-2025-124, which the reviewer classes as consistency. The commitment register shows consistency incident INC-2025-124 against an undertaking to annotate staleness on every read path and add a rebalance-under-load test to the platform suite, with Marta Kellerman named and 2025-11-30 committed. Robot Shadow Store incident INC-2025-124 is the only entry in its group where the reviewer and the engineering team reached different conclusions.

Strong on the cause and limited on the contributing factor is the evidence grade for observability event INC-2026-028, and the reader should not read severity into it. Fleet Health Monitor incident INC-2026-028 supports one conclusion and not another: an operational workaround masked the defect for longer than the incident record suggests. Fleet Health Monitor incident INC-2026-028 affected one site and 22 robots. Beyond the direct cause, Fleet Health Monitor incident INC-2026-028 shows that case closure did not check for unresolved constituent alerts. INC-2026-028 traces back to a single fact: alert grouping keyed on site and severity rather than on the underlying signal. Granite Peak Distribution incident INC-2026-028 was the subject of a specific management representation, which the reviewer has recorded but not relied on alone.

The remediation committed after Granite Peak Distribution incident INC-2026-028 was to group on the signal and block closure while a constituent alert remains active, owned by Wanda Njoku with a due date of 2026-04-30. Fleet Health Monitor accounts for Granite Peak Distribution incident INC-2026-028, a observability event. Fleet Health Monitor incident INC-2026-028 shows a short mitigation and a long tail: not a service outage; 2 days of delayed repair against 31 days. Observability incident INC-2026-028 was raised on 2026-03-10 after two distinct faults at one site were

grouped into a single case and the second went unworked for two days. The detection path for INC-2026-028 was the customer asked why one symptom persisted after the case had been closed. incident INC-2026-028 is the reason one section of TDR-2025-09 concludes narrowly rather than generally.

Schedule F - Incident population examined, with follow-through status (TDR-2025-09)

Incident	Date	Component	Sites affected	Robots affected	Time to mitigate	Time to resolve	Root cause class	Remediation landed
INC-2025-041	2025-03-14	Traffic Manager	2	61	1 h 52 m	9 days	resource leak	landed
INC-2025-047	2025-04-02	Lidar Driver	1	12	38 m	3 days	timing	partly landed
INC-2025-052	2025-04-21	Charge Scheduler	1	44	4 h 10 m	16 days	scheduling	landed
INC-2025-058	2025-05-09	WMS Connector	2	0	2 h 35 m	21 days	data integrity	landed
INC-2025-063	2025-05-27	Fleet Allocator	3	28	6 h 05 m	12 days	policy defect	landed
INC-2025-071	2025-06-11	Telemetry Ingest	6	214	52 m	52 m	availability	landed
INC-2025-078	2025-06-24	Remote Operations Console	4	0	1 h 15 m	34 days	stale state	landed
INC-2025-084	2025-07-08	OTA Updater	1	11	27 m	5 days	state persistence	landed
INC-2025-090	2025-07-19	Safety Controller	1	3	immediate stop, 22 m to return to service	41 days	hardware fault	partly landed
INC-2025-096	2025-08-05	Alarm Router	0	0	not applicable; no service impact	26 days	process gap	partly landed
INC-2025-103	2025-08-22	Motion Planner	1	2	19 m	58 days	algorithm defect	partly landed
INC-2025-110	2025-09-03	Analytics Warehouse Loader	1	0	not applicable; reporting only	18 days	data integrity	landed
INC-2025-118	2025-09-17	Identity and Authorization	0	0	3 h 40 m to revoke and rotate	7 days	secret exposure	partly landed
INC-2025-124	2025-10-01	Robot Shadow Store	5	178	1 h 08 m	29 days	consistency	landed
INC-2025-131	2025-10-14	Map Authoring Studio	1	34	3 h 22 m	44 days	process bypass	partly landed

2. INC-2025-090 and the incident population examined - part 2 of 2

Presentation of the 2. INC-2025-090 and the incident population examined carries on below without restarting the row numbers, the break being one of pagination only.

Whatever caveat the 2. INC-2025-090 and the incident population examined carried at its head travels with these rows, and no additional caveat attaches to this part. The break is editorial.

Schedule F - Incident population examined, with follow-through status (TDR-2025-09) (continued, part 2 of 2)

Incident	Date	Component	Sites affected	Robots affected	Time to mitigate	Time to resolve	Root cause class	Remediation landed
INC-2025-139	2025-11-02	Event Bus	7	246	2 h 47 m	63 days	capacity	partly landed
INC-2025-146	2025-11-19	Billing Usage Meter	0	0	not applicable; caught before billing	11 days	process defect	landed
INC-2025-152	2025-12-04	Video Snippet Service	0	0	not applicable; retention only	37 days	data retention	partly landed

Incident	Date	Component	Sites affected	Robots affected	Time to mitigate	Time to resolve	Root cause class	Remediation landed
INC-2026-004	2026-01-13	Localization Engine	1	18	not a service outage; 5 days to a workaround	open at the review date	performance	not landed
INC-2026-009	2026-01-27	Gripper Controller	1	4	1 h 30 m to quarantine the affected robots	48 days	calibration	landed
INC-2026-015	2026-02-09	Config Distribution	2	7	44 m	22 days	validation	landed
INC-2026-021	2026-02-24	Simulation Harness	0	0	not applicable; developer tooling	open at the review date	tooling	not landed
INC-2026-028	2026-03-10	Fleet Health Monitor	1	22	not a service outage; 2 days of delayed repair	31 days	observability	landed
INC-2026-033	2026-03-25	Speed and Separation Monitor	1	6	immediate speed reduction; 3 h to review the fleet	open at the review date	safety margin	not landed
INC-2026-039	2026-04-07	Order Broker	1	1	1 h 04 m	19 days	concurrency	landed
INC-2026-045	2026-04-21	Edge Gateway Agent	1	39	5 h 12 m	26 days	observability	partly landed
INC-2026-051	2026-05-06	Site Provisioning Service	1	0	2 h 18 m	9 days	configuration drift	partly landed
INC-2026-057	2026-05-19	Battery Management Service	2	5	not a service outage; robots derated	open at the review date	modelling	not landed
INC-2026-062	2026-06-02	Release Orchestrator	0	0	not applicable; process only	open at the review date	process control	not landed
INC-2026-068	2026-06-16	Wheel Odometry Filter	1	14	not a service outage; 3 h of degraded operation	open at the review date	environmental	not landed

3. Detection paths and time to mitigate

The two durations recorded for observability incident INC-2026-045 are 5 h 12 m to mitigate and 26 days to resolve. Attention came to Evergreen Warehousing incident INC-2026-045 when the buffer pressure metric was visible per robot but no site-level alert existed. Observability event INC-2026-045 traces back to a single fact: the buffer is sized for four hours and drops oldest-first without raising an alert. Following observability event INC-2026-045 the team undertook to aggregate buffer pressure to a site alert and extend the buffer to twelve hours on the current hardware, with Sam Oyelaran accountable by 2026-06-30. Evergreen Warehousing felt Edge Gateway Agent incident INC-2026-045, which affected one site and 39 robots. Edge Gateway Agent incident INC-2026-045 is the clearest instance in TDR-2025-09 of engineering practice being better than the documentation suggests.

Taken with the rest of the record, observability incident INC-2026-045 indicates that the failure was in the detection design rather than in the code that failed. Grading INC-2026-045 on evidence rather than on consequence gives strong on the cause and limited on the contributing factor. Verification of the observability event INC-2026-045 remediation gives the alert landed on 2026-05-17; the buffer extension is in progress. The trigger recorded for INC-2026-045 on 2026-04-21 was that a site-wide network change left robots buffering for five hours and the oldest telemetry samples were discarded. INC-2026-045 carries a second-order finding: buffer pressure was never aggregated to a site view, so the condition was not observable. Edge Gateway Agent incident INC-2026-045 was inspected in the repository rather than described from the ticket, which is why the account is specific.

Tooling event INC-2026-021 is recorded as follows on delivery: not landed; the determinism check is written but not enabled. Colin Achebe committed after Simulation Harness incident INC-2026-021 to pin the full build configuration and add a determinism check to the harness, due 2026-05-31. Incident INC-2026-021 touched no production site and no robot directly, the exposure being to internal systems, and internal was the affected account. Underneath INC-2026-021 lies the design decision that the physics core is not deterministic across optimisation levels. Detection came through the autonomy team noticed a step change in the pass rate, which places tooling incident INC-2026-021 on the reported side of the ledger. incident INC-2026-021 illustrates why TDR-2025-09 separates the severity of a finding from the confidence behind it.

Tooling event INC-2026-021 was mitigated in not applicable; developer tooling; the resolution ran to open at the review date. 2026-02-24 is the date of Simulation Harness incident INC-2026-021, on which a nightly simulation suite produced non-reproducible results after the physics core was rebuilt. The contributing factor recorded against incident INC-2026-021 is that the build pinned the compiler but not the optimisation flags used for release builds. Classifying Simulation Harness incident INC-2026-021 as tooling matters because the Simulation Harness backlog contains two further items of the same kind. The reviewer's reading of incident INC-2026-021 is that the blast radius was set by a deployment topology decision made years earlier. tooling incident INC-2026-021 sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

After calibration incident INC-2026-009, Ines Duarte took the action to require a calibration record before returning a manipulator robot to service by 2026-04-15. Calibration incident INC-2026-009 touched one site and 4 robots, and Blue Mesa Components was the affected account. Calibration incident INC-2026-009 records that on 2026-01-27, a gripper applied more force than the configured limit after a mechanical replacement without recalibration. Blue Mesa Components incident INC-2026-009 is recorded as follows on delivery: landed on 2026-03-16 with a workflow gate. Duration for incident INC-2026-009 splits into 1 h 30 m to quarantine the affected robots of impact and 48 days to close the underlying cause. calibration incident INC-2026-009 would move from a concern to a verified defect if one further measurement were supplied.

Gripper Controller incident INC-2026-009 is not read as an isolated failure: the customer-reported detection path is the finding, not the defect itself. The engineering team's root cause for INC-2026-009 is that force limiting depends on a calibration that is only triggered by a service workflow step, and the reviewer found the analysis sound. Blue Mesa Components incident INC-2026-009 carries a second-order finding: the service workflow allowed the step to be marked complete without a calibration record. The reviewer grades the calibration incident INC-2026-009 evidence as strong, resting on the production traces the team supplied and has not upgraded the finding to compensate. Gripper Controller accounts for calibration incident INC-2026-009, a calibration event. incident INC-2026-009 is the only entry in its group where the reviewer and the engineering team reached different conclusions.

Incident INC-2025-047 would have been shorter or smaller had it not been that the bench rig that would have caught it carries a different lidar variant. For timing event INC-2025-047 the reviewer holds moderate; the narrative is consistent but the reviewer could not independently reproduce it evidence and reports the finding at that strength. The engineering team's root cause for INC-2025-047 is that the driver did not re-derive the sensor time offset on reconnect, and the reviewer found the analysis sound. Timing event INC-2025-047 took 38 m to stabilise and 3 days to close out. INC-2025-047 was raised on 2025-04-02 after a lidar unit reconnected after a cable seating fault and reported frames with a stale timestamp offset. Asterion Logistics incident INC-2025-047 was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

What Lidar Driver incident INC-2025-047 tells the reviewer is that management's severity rating is defensible and its remediation priority is not. Incident INC-2025-047 reached the engineering team when an automated fusion timing alert fired inside four minutes. In taxonomy terms timing event INC-2025-047 is a timing incident in Lidar Driver. Asterion Logistics felt Lidar Driver incident INC-2025-047, which affected one site and 12 robots. Remediation for Asterion Logistics incident INC-2025-047 is to re-derive the time offset on every reconnect and add the affected variant to the second bench rig; Yuki Harada owns it and the committed date is 2025-06-15. INC-2025-047 was compared with the equivalent position at two other components, and it is neither the best nor the worst.

The engineering team's root cause for INC-2026-062 is that the emergency path is available to any release engineer without a justification field, and the reviewer found the analysis sound. In INC-2026-062 the first signal was the release

audit found it during the quarterly review, and the reviewer treats that as reported detection. On 2026-06-02 the record for process control incident INC-2026-062 opens: an emergency release path with a single approver was used for a change that was not an emergency. INC-2026-062 supports one conclusion and not another: the remediation addresses the immediate cause and leaves the contributing factor in place. Process control is the class of Release Orchestrator incident INC-2026-062; Release Orchestrator is the component. process control event INC-2026-062 is not material on its own and belongs in this report because of what it reveals about the surrounding process.

In Release Orchestrator incident INC-2026-062, the policy defines an emergency but the tool does not ask which condition applies, which is a process finding rather than a code finding. Not landed; the change is scoped but not scheduled is the position on the Release Orchestrator incident INC-2026-062 commitment as at the review date. Following Release Orchestrator incident INC-2026-062 the team undertook to require a justification and a second approver within one working day for every emergency release, with Colin Achebe accountable by 2026-08-31. Evidence strength on the INC-2026-062 analysis is moderate, resting on the team's own post-mortem and one supporting log extract. Duration for incident INC-2026-062 splits into not applicable; process only of impact and open at the review date to close the underlying cause. incident INC-2026-062 is one of the entries where TDR-2025-09 relies on the company's own measurement rather than an independent one.

Blue Mesa Components incident INC-2025-084 closure status is landed on 2025-07-14. Following INC-2025-084 the team undertook to persist the pause state and require a design note for any change to the release state machine, with Erik Palmgren accountable by 2025-08-15. State persistence incident INC-2025-084 took 27 m to stabilise and 5 days to close out. Incident INC-2025-084 classifies as a state persistence event against OTA Updater. How Blue Mesa Components incident INC-2025-084 surfaced matters to the assessment: the site team escalated when robots rebooted outside the maintenance window. Blue Mesa Components incident INC-2025-084 illustrates why TDR-2025-09 separates the severity of a finding from the confidence behind it.

Drawing the evidence in OTA Updater incident INC-2025-084 together, the customer-reported detection path is the finding, not the defect itself. Reading incident INC-2025-084 from its first entry, on 2025-07-08 a paused update ring resumed after a routine service restart and eleven robots took an unintended firmware step. For Blue Mesa Components incident INC-2025-084 the reviewer holds moderate on the cause and strong on the delivery position, which was checked in the repository evidence and reports the finding at that strength. Blue Mesa Components incident INC-2025-084 was caused by a design gap: the pause flag was held in process memory rather than in the release state record. Blue Mesa Components incident INC-2025-084 carries a second-order finding: the pause control had been added in a hotfix without a design review. state persistence incident INC-2025-084 would be a candidate for the first post-close engineering review under any reasonable plan.

Assessed on its own record, INC-2025-118 shows that the blast radius was set by a deployment topology decision made years earlier. The detection path for internal incident INC-2025-118 was the dependency and secret scanner flagged it on the following scan. Following secret exposure event INC-2025-118 the team undertook to redact authorization headers at the logging layer and cut build log retention to fourteen days, with Sonia Krall accountable by 2025-10-31. Underneath internal incident INC-2025-118 lies the design decision that a debug logging statement in the connector build printed the full authorization header. Secret exposure incident INC-2025-118 shows a short mitigation and a long tail: 3 h 40 m to revoke and rotate against 7 days. incident INC-2025-118 was checked against the corresponding entry in the company's own register, and the register was current.

Classifying incident INC-2025-118 as secret exposure matters because the Identity and Authorization backlog contains two further items of the same kind. The reviewer grades the internal incident INC-2025-118 evidence as strong, since the reviewer reproduced the failure mode in a laboratory environment and has not upgraded the finding to compensate. The trigger recorded for INC-2025-118 on 2025-09-17 was that a service token with a twelve-hour lifetime was found in a build log retained for ninety days. INC-2025-118 would have been shorter or smaller had it not been that build logs were retained far longer than the token lifetime, widening the exposure window. Secret exposure incident INC-2025-118 was contained to no production site and no robot directly, the exposure being to internal systems. INC-2025-118 would move from a concern to a verified defect if one further measurement were supplied.

Algorithm defect event INC-2025-103 began on 2025-08-22 when two robots repeatedly yielded to each other in a narrow aisle and both stalled for nineteen minutes. Priya Raghunathan committed after Motion Planner incident

INC-2025-103 to add a deterministic tie-break and return the pinch-point scenario to the nightly suite, due 2025-10-31. Motion Planner incident INC-2025-103 was mitigated in 19 m; the resolution ran to 58 days. Algorithm defect incident INC-2025-103 rests on moderate, resting on the team's own post-mortem and one supporting log extract evidence, which the reviewer states separately from the severity of the finding. Cause analysis in incident INC-2025-103 concluded that the yield heuristic had no tie-break and both robots computed the same deference score. Motion Planner incident INC-2025-103 was tested rather than accepted, and the test is described in the working papers behind TDR-2025-09.

Algorithm defect event INC-2025-103 is not read as an isolated failure: the commitment date was met and the verifying test now exists, which is not the pattern across the whole register. In Ironwood Grocery Group incident INC-2025-103 the first signal was the queue depth alert introduced after the March incident fired correctly, and the reviewer treats that as alerted detection. Algorithm defect event INC-2025-103 is recorded as follows on delivery: the tie-break landed on 2025-10-19; the scenario was reinstated on 2025-09-02. The affected population in algorithm defect incident INC-2025-103 was one site and 2 robots. A contributing factor in Motion Planner incident INC-2025-103 was that the simulation scenario for this geometry existed but had been excluded from the nightly suite. algorithm defect event INC-2025-103 has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

4. Root causes, contributing factors and recurrence risk

Detection came through a robot derated mid-shift and the pack was pulled for analysis, which places incident INC-2026-057 on the reported side of the ledger. Cause analysis in INC-2026-057 concluded that the estimator was fitted on packs that follow a full charge cycle. Modelling event INC-2026-057 carries a second-order finding: the two affected sites moved to opportunity charging after the estimator had been validated. Modelling incident INC-2026-057 is not read as an isolated failure: the same class of defect can recur in two sibling components that were not changed. Following Battery Management Service incident INC-2026-057 the team undertook to refit the estimator on opportunity-charge data and add an imbalance alert independent of state of health, with Peter Halvorsen accountable by 2026-09-30. Battery Management Service incident INC-2026-057 is treated as material to the investment case rather than as a routine engineering item.

Modelling incident INC-2026-057 shows a short mitigation and a long tail: not a service outage; robots derated against open at the review date. Modelling incident INC-2026-057 was contained to 2 sites and 5 robots. For modelling event INC-2026-057 the reviewer holds moderate, resting on the team's own post-mortem and one supporting log extract evidence and reports the finding at that strength. The trigger recorded for incident INC-2026-057 on 2026-05-19 was that state-of-health estimates on opportunity-charged packs drifted enough to hide a cell imbalance until derating. Verification of the INC-2026-057 remediation gives not landed; data collection is running and the imbalance alert is in review. modelling incident INC-2026-057 was raised again with the owning engineer after the initial interview, and the position did not change.

The reviewer treats one contributing factor in process gap event INC-2025-096 as the more important lesson: the rotation source of truth was a spreadsheet maintained outside the alarm router. Strong, resting on the production traces the team supplied is the evidence grade for process gap event INC-2025-096, and the reader should not read severity into it. What went wrong in internal incident INC-2025-096 was that the routing table fell back to a mailbox rather than to a secondary responder. Not applicable; no service impact elapsed before incident INC-2025-096 was mitigated, with resolution at 26 days. Process gap incident INC-2025-096 affected no production site and no robot directly, the exposure being to internal systems. Alarm Router incident INC-2025-096 was raised again with the owning engineer after the initial interview, and the position did not change.

The classification applied to process gap event INC-2025-096 is process gap, and the affected component is Alarm Router. The detection path for incident INC-2025-096 was discovered the next morning when the alarm was found unacknowledged. Alarm Router incident INC-2025-096 produced one commitment: to replace the mailbox fallback with a secondary responder and import rotations from the scheduling system, owned by Wanda Njoku, dated 2025-10-15. On 2025-08-05 the record for internal incident INC-2025-096 opens: an overnight alarm routed to a shared mailbox because the on-call rotation had a two-hour gap. The reviewer checked the incident INC-2025-096 commitment against the code and release records; the fallback change landed on 2025-08-31; the rotation import was deferred to 2026. INC-2025-096 has no single accountable owner, and that is the finding the reviewer would press hardest.

Performance event INC-2026-004 touched one site and 18 robots, and Kestrel Medical Supply was the affected account. In incident INC-2026-004, the site was surveyed with a sparse feature set that increases cache misses, which is a process finding rather than a code finding. Attention came to Kestrel Medical Supply incident INC-2026-004 when the site's commissioning engineer raised it during acceptance testing. The corrective action for Localization Engine incident INC-2026-004 is to replace the fallback matcher and re-survey the affected aisles, and 2026-06-30 is the date the team gave. As to whether the Kestrel Medical Supply incident INC-2026-004 remediation actually landed: not landed; the matcher replacement is in design and the re-survey is scheduled. Kestrel Medical Supply incident INC-2026-004 rests on a single source, and the reviewer has said so rather than implying corroboration.

What went wrong in performance event INC-2026-004 was that the fallback scan matcher is slower than the primary and is selected whenever the map cache misses. Localization Engine incident INC-2026-004 was raised on 2026-01-13 after relocalisation after a lift-and-place exceeded the operator prompt timeout at a newly surveyed site. INC-2026-004 shows a short mitigation and a long tail: not a service outage; 5 days to a workaround against open at the review date. Localization Engine accounts for performance event INC-2026-004, a performance event. The reviewer grades the incident INC-2026-004 evidence as limited, being a single account corroborated only by the ticket timeline and has not upgraded the finding to compensate. Localization Engine incident INC-2026-004 is the reason one section of TDR-2025-09 concludes narrowly rather than generally.

Drawing the evidence in multiple sites incident INC-2025-139 together, management's severity rating is defensible and its remediation priority is not. Multiple sites incident INC-2025-139 touched 7 sites and 246 robots, and multiple sites was the affected account. The post-mortem for INC-2025-139 identifies the cause as follows: control and telemetry traffic shared one topic and one consumer group budget. Moderate; the narrative is consistent but the reviewer could not independently reproduce it is the evidence grade for INC-2025-139, and the reader should not read severity into it. Capacity incident INC-2025-139 was mitigated in 2 h 47 m; the resolution ran to 63 days. capacity event INC-2025-139 has no single accountable owner, and that is the finding the reviewer would press hardest.

INC-2025-139 classifies as a capacity event against Event Bus. The immediate event in incident INC-2025-139 was that a telemetry backlog on a shared topic delayed control events by up to eleven seconds, dated 2025-11-02. Following INC-2025-139 the team undertook to split control traffic onto a dedicated topic with its own quota and revisit the topic inventory quarterly, with Erik Palmgren accountable by 2026-01-31. Event Bus incident INC-2025-139 was detected because the traffic manager's own latency alert fired first. On delivery, capacity incident INC-2025-139 shows this: the split landed on 2026-01-04; the quarterly review has not yet run. incident INC-2025-139 rests on a single source, and the reviewer has said so rather than implying corroboration.

Incident INC-2025-041 closure status is landed on 2025-05-22, verified in the reservation replay suite. For resource leak incident INC-2025-041 the reviewer holds moderate; the narrative is consistent but the reviewer could not independently reproduce it evidence and reports the finding at that strength. The trigger recorded for incident INC-2025-041 on 2025-03-14 was that a robot lost its network link mid-traverse and its corridor reservation was never released. INC-2025-041 traces back to a single fact: the reservation lease had no expiry, so release depended on the departing robot. The detection path for incident INC-2025-041 was a site supervisor telephoned the on-call engineer after robots queued at one intersection. incident INC-2025-041 was checked against the corresponding entry in the company's own register, and the register was current.

Duration for Traffic Manager incident INC-2025-041 splits into 1 h 52 m of impact and 9 days to close the underlying cause. Crestline Fulfillment felt Crestline Fulfillment incident INC-2025-041, which affected 2 sites and 61 robots. Traffic Manager incident INC-2025-041 produced one commitment: to give every corridor reservation a lease with an expiry and add a site-level queue depth alert, owned by Owen Brantley, dated 2025-05-30. Traffic Manager incident INC-2025-041 carries a second-order finding: the alert rule for queue depth existed but was scoped to a metric the site did not emit. Resource leak event INC-2025-041 classifies as a resource leak event against Traffic Manager. Traffic Manager incident INC-2025-041 was checked against the corresponding entry in the company's own register, and the register was current.

Stale state event INC-2025-078 was detected because an internal engineer noticed the discrepancy during a demonstration. INC-2025-078 shows a short mitigation and a long tail: 1 h 15 m against 34 days. The corrective action for incident INC-2025-078 is to invalidate the roster cache on a fleet membership event and add a session-level regression

test, and 2025-08-31 is the date the team gave. Incident INC-2025-078 carries strong, resting on the production traces the team supplied evidence, and the reviewer declined to raise the finding beyond what that supports. Beyond the direct cause, stale state incident INC-2025-078 shows that no test covered a roster change during an open operator session. stale state event INC-2025-078 is reported at the evidence strength available and has not been promoted to fit the narrative.

The reviewer's reading of stale state event INC-2025-078 is that the incident is unremarkable in itself and revealing about the observability of the surrounding system. Root cause in Remote Operations Console incident INC-2025-078 was that the session list cached the site roster and only invalidated it on a page reload. Stale state incident INC-2025-078 was contained to 4 site or tenant with no robot taken out of service. Stale state event INC-2025-078 was raised on 2025-06-24 after the console listed robots that had been removed from a site and an operator commanded a decommissioned unit. Classifying stale state incident INC-2025-078 as stale state matters because the Remote Operations Console backlog contains two further items of the same kind. INC-2025-078 is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

Schedule G - Open defect population and ageing by service (TDR-2025-09)

Service	Open defects	Oldest open item	Severity mix	Closed in reviewed period	Engineering owner
Motion Planner	14	2025-12-06	1 high, 2 medium, 11 low	2	Priya Raghunathan
Costmap Fusion	21	2026-02-24	0 high, 2 medium, 19 low	3	Daniel Okafor
Fleet Allocator	9	2025-06-04	2 high, 2 medium, 5 low	3	Marta Kellerman
Traffic Manager	12	2026-01-13	1 high, 3 medium, 8 low	12	Owen Brantley
Charge Scheduler	7	2025-08-13	0 high, 1 medium, 6 low	7	Nadia Semenova
Safety Controller	3	2025-10-06	2 high, 1 medium, 0 low	5	Bettina Vogel
E-Stop Chain	2	2025-09-15	2 high, 1 medium, 0 low	13	Theo Bassett
Lidar Driver	18	2026-02-22	2 high, 2 medium, 14 low	11	Yuki Harada
Localization Engine	16	2025-08-10	0 high, 1 medium, 15 low	1	Rahul Menon
WMS Connector	23	2025-07-15	0 high, 2 medium, 21 low	4	Tomas Villareal
Order Broker	11	2026-04-21	1 high, 2 medium, 8 low	7	Grace Lindqvist
Telemetry Ingest	19	2026-03-04	2 high, 5 medium, 12 low	14	Farida Cisse
OTA Updater	8	2025-04-09	1 high, 3 medium, 4 low	11	Erik Palmgren
Identity and Authorization	6	2025-07-30	0 high, 1 medium, 5 low	11	Sonia Krall
Remote Operations Console	27	2025-03-03	2 high, 3 medium, 22 low	10	Anjali Sekhon
Simulation Harness	31	2026-03-09	1 high, 1 medium, 29 low	13	Colin Achebe
Hardware-in-the-Loop Rig	13	2026-03-01	0 high, 2 medium, 11 low	11	Nathan Oduya

4. Root causes, contributing factors and recurrence risk - part 2 of 2

Presentation of the 4. Root causes, contributing factors and recurrence risk carries on below without restarting the row numbers, the break being one of pagination only. Pagination alone determines where it falls.

Readers should carry the 4. Root causes, contributing factors and recurrence risk's stated conventions across the break, since this part adds no definition of its own. No row is omitted at the join.

Schedule G - Open defect population and ageing by service (TDR-2025-09) (continued, part 2 of 2)

Service	Open defects	Oldest open item	Severity mix	Closed in reviewed period	Engineering owner
Fleet Health Monitor	17	2025-08-22	2 high, 4 medium, 11 low	1	Wanda Njoku

Service	Open defects	Oldest open item	Severity mix	Closed in reviewed period	Engineering owner
Battery Management Service	10	2026-04-19	1 high, 5 medium, 4 low	7	Peter Halvorsen
Gripper Controller	15	2025-05-30	2 high, 4 medium, 9 low	11	Ines Duarte
Map Authoring Studio	22	2025-10-14	2 high, 10 medium, 10 low	6	Lucia Ferrante
Site Provisioning Service	5	2025-04-01	1 high, 2 medium, 2 low	5	Marcus Duplessis
Event Bus	4	2025-10-22	2 high, 2 medium, 0 low	9	Erik Palmgren
Config Distribution	6	2026-01-14	1 high, 1 medium, 4 low	4	Sam Oyelaran
Robot Shadow Store	9	2025-10-13	2 high, 1 medium, 6 low	6	Marta Kellerman
Analytics Warehouse Loader	20	2025-10-06	1 high, 3 medium, 16 low	3	Farida Cisse
Billing Usage Meter	7	2025-09-05	0 high, 1 medium, 6 low	11	Nadia Semenova
Alarm Router	14	2025-10-09	0 high, 4 medium, 10 low	4	Wanda Njoku
Video Snippet Service	12	2025-09-19	0 high, 6 medium, 6 low	11	Anjali Sekhon
Speed and Separation Monitor	4	2026-03-26	0 high, 1 medium, 3 low	5	Bettina Vogel
Wheel Odometry Filter	8	2026-03-29	0 high, 1 medium, 7 low	13	Yuki Harada
Payload Scale Interface	11	2026-04-01	0 high, 2 medium, 9 low	2	Ines Duarte
Edge Gateway Agent	5	2025-11-16	1 high, 2 medium, 2 low	5	Sam Oyelaran
Release Orchestrator	16	2025-09-09	0 high, 5 medium, 11 low	14	Colin Achebe

5. Reliability of the services carrying the incidents

Wheel Odometry Filter was checked for shared-database coupling, and the ownership boundary held. Coverage for Wheel Odometry Filter was 62 per cent line coverage at the review date, and the trend over the reviewed period was up by four points. The manifests place Wheel Odometry Filter edge. Wheel Odometry Filter releases with the firmware, twice a year. Wheel Odometry Filter carries 8 open defects, among them a case where drift exceeds the specification on wet or dusty floors and is not flagged to the operator. Wheel Odometry Filter is one of the entries where TDR-2025-09 relies on the company's own measurement rather than an independent one.

Underlying Wheel Odometry Filter in this report are the filter design note, drift measurements at four sites and a wet-floor bench run. Wheel Odometry Filter is written in C++17, executes on on-robot Linux, in the drive control process, and keeps its state in no persistent store. The objective recorded against Wheel Odometry Filter is that odometry drift stays under 1.5 per cent of distance travelled on a clean floor, and it is reported weekly to the engineering leadership. The reviewer's judgement on Wheel Odometry Filter is that the component is adequate today and carries a supportability liability that is not on the funded plan. On end-of-life exposure, Wheel Odometry Filter shows none identified. Wheel Odometry Filter was checked against the corresponding entry in the company's own register, and the register was current.

Coverage for Localization Engine was 66 per cent line coverage at the review date, and the trend over the reviewed period was down by two points after a large refactor. The dependency map places the lidar driver, the wheel odometry filter and the map distribution channel above Localization Engine and the motion planner and the fleet health monitor below it. The interface Localization Engine publishes is versioned, and the motion planner and the fleet health monitor were checked for use of any undocumented field. C++17 with a Rust bridge layer is the implementation language for Localization Engine, which is hosted on on-robot Linux, pinned to a dedicated core with a local map cache with a periodic checkpoint behind it. Documents and data inspected for Localization Engine were the localization design note, survey reports from three sites and the reviewer's map audit. Localization Engine is reported here in full because a summary of it in an earlier draft of TDR-2025-09 was misread as an assurance.

The ageing position for Localization Engine is an unmaintained scan-matching library is still linked in the fallback path, and the reviewer checked it against the vendor statements on file. The Autonomy Core team carries Localization Engine on its service inventory, and Rahul Menon answered for it in interview. Localization Engine runs edge, with map distribution from the cloud; that placement is what determines whether a fault is one robot or one region. Service-level intent for Localization Engine states that pose error stays under 25 mm in mapped aisles at survey speed. Against Localization Engine the backlog runs to 16 items, and it includes a known behaviour whereby relocalisation after a lift-and-place takes longer than the operator prompt allows. Localization Engine appears twice in the working papers for TDR-2025-09 because the first answer was corrected after a follow-up request.

Map Authoring Studio was assessed against the objective and the defect record, and management's own assessment is broadly supported, with the coverage gap treated as a genuine finding rather than a presentational one. The cadence recorded for Map Authoring Studio is weekly, which matches the observed release history. Map Authoring Studio was assessed on the authoring workflow document, the revision history for six sites and an export audit, and nothing beyond that was available in the data room. TypeScript with a Python service layer is the implementation language for Map Authoring Studio, which is hosted on cloud, a desktop-class web application with PostgreSQL 14 with map revisions in object storage behind it. Applications tracks 22 open defects on Map Authoring Studio, including the acknowledged case where an unpublished draft can be exported and hand-loaded, bypassing the review step. Map Authoring Studio was reviewed against the position twelve months earlier, and the direction of travel is favourable.

Map Authoring Studio sits at 49 per cent line coverage; the trend across the reviewed period was flat. Service-level intent for Map Authoring Studio states that a published map revision reaches every robot at the site inside fifteen minutes. Lucia Ferrante of Applications maintains Map Authoring Studio, and the service inventory entry was current at the review date. The interface Map Authoring Studio publishes is versioned, and the map distribution channel and the localization engine were checked for use of any undocumented field. Supportability of Map Authoring Studio turns on one fact: a geometry rendering library one major version behind its supported line. Map Authoring Studio is reported at the evidence strength available and has not been promoted to fit the narrative.

31 items sit in the Simulation Harness backlog; the entry that bears on the investment case is that scenario determinism breaks when the physics core is rebuilt with a different optimisation level. Simulation Harness rests on the scenario catalogue, two weeks of nightly run records and a determinism experiment so far as this review is concerned. Reading the dependency map from Simulation Harness outward gives the planner and localization builds under test on the input side and the release orchestrator gate and the autonomy regression report on the output side. Behind Simulation Harness sits object storage for scenario definitions and run artefacts; the service itself is Python 3.11 with a C++ physics core on developer workstations and a batch cluster. Simulation Harness reports 47 per cent line coverage, measured by the same instrumentation the pipeline uses. Simulation Harness is the clearest instance in TDR-2025-09 of a control that exists on paper and is weaker in practice.

Ownership of Simulation Harness sits with Developer Productivity; Colin Achebe signs off changes to it. Simulation Harness runs cloud batch, with a local mode for developers; that placement is what determines whether a fault is one robot or one region. Where Simulation Harness is concerned, the commitment is that the nightly scenario suite completes inside four hours; the reviewer traced the metric to its source. Supportability of Simulation Harness turns on one fact: the physics core depends on a library branch that stopped receiving fixes in 2023. Simulation Harness was checked for shared-database coupling, and the ownership boundary held. Simulation Harness was excluded from an earlier draft of TDR-2025-09 and reinstated after the reviewer obtained the supporting record.

Underlying E-Stop Chain in this report are the wiring diagrams, the change order file and a witnessed actuation test on four units. Safety Engineering tracks 2 open defects on E-Stop Chain, including the acknowledged case where one pre-production harness revision remains in the field on nine robots at two sites. Hosting for E-Stop Chain is a safety-rated hardware assembly, no general-purpose operating system, and its VHDL with discrete relay logic code writes through to no persistent store; events are latched to the safety controller. E-Stop Chain was weighed on its coverage, its defect backlog and its dependency position, and the risk is concentrated in one code path that has neither automated coverage nor a bench equivalent. E-Stop Chain sits at no meaningful automated unit coverage, verification being by witnessed test instead; the trend across the reviewed period was down by two points after a large refactor. E-Stop Chain was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

On end-of-life exposure, E-Stop Chain shows none; the relay parts are current production. Reading the dependency map from E-Stop Chain outward gives the operator stop buttons and the perimeter interlocks on the input side and the safety controller and the drive contactors on the output side. Safety Engineering holds E-Stop Chain, and the ownership entry naming Theo Bassett matches the on-call rotation. E-Stop Chain owns its own data and exposes it only through a versioned interface, which the reviewer read. The topology for E-Stop Chain is edge, hardwired, independent of the compute module, and the reviewer confirmed it against the deployment manifests. E-Stop Chain has no single accountable owner, and that is the finding the reviewer would press hardest.

Hardware-in-the-Loop Rig's dependency ageing was checked: the rig controller runs an operating system release that left vendor support in 2025. Hardware-in-the-Loop Rig publishes a stable interface, though two fields used by the firmware release decision record are documented only in code comments. 39 per cent line coverage is reported for Hardware-in-the-Loop Rig, and the reviewer re-ran the measurement on a clean checkout. Responsibility for Hardware-in-the-Loop Rig was confirmed with Nathan Oduya, whose team, Robot Firmware, also holds the on-call duty. The backlog for Hardware-in-the-Loop Rig is 13 items deep and contains an acknowledged defect whereby the second rig carries a superseded lidar variant, so one sensor path is only tested on one rig. Hardware-in-the-Loop Rig was reviewed against the position twelve months earlier, and nothing measurable has changed.

Where Hardware-in-the-Loop Rig is concerned, the commitment is that every firmware candidate runs the full bench suite before ring one; the reviewer traced the metric to its source. Hardware-in-the-Loop Rig was weighed on its coverage, its defect backlog and its dependency position, and the design is sound and the test evidence is thinner than the design deserves. Placement of Hardware-in-the-Loop Rig is laboratory, two rigs, one of which lags the current sensor configuration, which the reviewer tested by tracing a request end to end. Hardware-in-the-Loop Rig is promoted the harness changes with each firmware release, with the ring order fixed by policy. Hardware-in-the-Loop Rig was assessed on the bench suite definition, rig configuration inventory and two firmware release records, and nothing beyond that was available in the data room. Hardware-in-the-Loop Rig was checked against the corresponding entry in the company's own register, and the register was current.

On paper Motion Planner releases every three weeks, promoted ring by ring, and the observed practice did not diverge from that. Motion Planner sits between the costmap fusion pipeline and the localization engine and the traffic manager and the on-robot drive supervisor, and that position is what makes its availability objective load-bearing. For Motion Planner the reviewer worked from eleven design records, the planner benchmark suite and four weeks of production plan traces. The manifests place Motion Planner edge, one instance per robot with no cloud fallback. Test coverage on Motion Planner is 71 per cent line coverage and the measurement basis was confirmed with Priya Raghunathan. Motion Planner was tested rather than accepted, and the test is described in the working papers behind TDR-2025-09.

For Motion Planner the supportability position is none; the toolchain is pinned to a supported compiler release. The assessment of Motion Planner is not unqualified: no verified defect was identified, and two plausible concerns are recorded below at low evidence strength. Motion Planner owns its own data and exposes it only through a versioned interface, which the reviewer read. Motion Planner has a single named owner, Priya Raghunathan of Autonomy Core, which is not true of every service reviewed. Motion Planner carries a single measurable objective, namely that plan cycle completes inside 40 ms at the 99th percentile. Motion Planner would move from a concern to a verified defect if one further measurement were supplied.

Schedule H - Change volume, change failure rate and restore time by month (TDR-2025-09)

Month	Changes to production	Failed changes	Change failure rate	Mean time to restore	Rollbacks
2025-01	66	2	3.0%	59 min	0
2025-02	46	4	8.6%	87 min	2
2025-03	63	2	3.1%	39 min	1
2025-04	83	3	3.6%	49 min	1
2025-05	61	1	1.6%	71 min	1
2025-06	58	5	8.6%	62 min	3
2025-07	59	5	8.4%	52 min	0
2025-08	54	0	0.0%	30 min	1
2025-09	44	0	0.0%	21 min	1
2025-10	75	4	5.3%	31 min	2

Month	Changes to production	Failed changes	Change failure rate	Mean time to restore	Rollbacks
2025-11	51	1	1.9%	81 min	0
2025-12	70	4	5.7%	80 min	1
2026-01	79	5	6.3%	73 min	2
2026-02	80	6	7.5%	42 min	3
2026-03	60	2	3.3%	56 min	2
2026-04	56	1	1.7%	54 min	1
2026-05	51	2	3.9%	49 min	3
2026-06	39	1	2.5%	77 min	1
2026-07	57	0	0.0%	80 min	0
2026-08	75	0	0.0%	64 min	0

6. Where the incidents landed: sites and customers

Chino, California reports 54% headroom against its integration with vendor D. What distinguishes Chino, California is that it is the go-live delayed by the forked provisioning template. Chino, California appears twice in the working papers for TDR-2025-09 because the first answer was corrected after a follow-up request.

Capacity headroom at Edwardsville, Illinois is 63%, and the warehouse system is vendor B. 2026-03 is the go-live month at Edwardsville, Illinois, where Crestline Fulfillment now runs 24 robots. Edwardsville, Illinois is included because a reader of TDR-2025-09 should be able to disagree with the inference without repeating the work.

Lehigh Valley, Pennsylvania was included in the sample because it is commissioned during the connector rework. Site ST-05 is Lehigh Valley, Pennsylvania: Crestline Fulfillment, 64 robots, live from 2024-02. Lehigh Valley, Pennsylvania is treated as material to the investment case rather than as a routine engineering item.

On Kenosha, Wisconsin the record shows that it is the second opportunity-charging site. Kenosha, Wisconsin reports 49% headroom against its integration with vendor D. Kenosha, Wisconsin was raised again with the owning engineer after the initial interview, and the position did not change.

Whitestown, Indiana is the site with the standing map publication workaround, and the reviewer weighted the sample accordingly. 44 robots are deployed at Whitestown, Indiana, which went live in 2025-09. Whitestown, Indiana is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

Headroom at Reno, Nevada stands at 57%; vendor D is the integrated warehouse system. Site ST-11 is Reno, Nevada: Blue Mesa Components, 29 robots, live from 2025-05. Reno, Nevada appears twice in the working papers for TDR-2025-09 because the first answer was corrected after a follow-up request.

Against its peak profile, Groveport, Ohio retains 44% headroom on a vendor A integration. For Groveport, Ohio the material fact is that it is opportunity charging only, which is where the pack estimator drift appeared. Groveport, Ohio was the subject of a specific management representation, which the reviewer has recorded but not relied on alone.

At Tracy, California the fleet is 58 robots, and the site has been live since 2025-01. 36% of headroom remains at Tracy, California, whose connector targets vendor B. Tracy, California was tested rather than accepted, and the test is described in the working papers behind TDR-2025-09.

At Newnan, Georgia the fleet is 66 robots, and the site has been live since 2025-07. Remaining capacity at Newnan, Georgia is 33%, measured against the site's peak order profile. Newnan, Georgia is the reason one section of TDR-2025-09 concludes narrowly rather than generally.

Site ST-18 is Mesquite, Texas: Evergreen Warehousing, 16 robots, live from 2026-06. Remaining capacity at Mesquite, Texas is 71%, measured against the site's peak order profile. Mesquite, Texas is included because a reader of TDR-2025-09 should be able to disagree with the inference without repeating the work.

Schedule J - Site deployments, capacity headroom and integration target (TDR-2025-09)

Site	Customer	Robots deployed	Go-live	Capacity headroom	Warehouse system	Note
Fremont, California	internal engineering site	34	2021-06	not measured	test harness	the reference site, and the only one with a laboratory alongside
Meridian, Idaho	Asterion Logistics	96	2022-09	22%	vendor A	the longest-running customer

Site	Customer	Robots deployed	Go-live	Capacity headroom	Warehouse system	Note
						deployment and the source of the most incident history
Grand Prairie, Texas	Asterion Logistics	78	2023-04	31%	vendor A	the second Asterion site, commissioned on the same template
Wilmington, Ohio	Crestline Fulfillment	112	2023-08	14%	vendor B	the largest deployment and the closest to the allocator scaling limit
Lehigh Valley, Pennsylvania	Crestline Fulfillment	64	2024-02	38%	vendor B	commissioned during the connector rework
Ontario, California	Evergreen Warehousing	88	2024-05	26%	vendor C	the site whose shift pattern change caused the April 2025 charge incident
Groveport, Ohio	Granite Peak Distribution	52	2024-09	44%	vendor A	opportunity charging only, which is where the pack estimator drift appeared
Kenosha, Wisconsin	Dunbar Industrial	41	2024-11	49%	vendor D	the second opportunity-charging site
Tracy, California	Harborline Systems	58	2025-01	36%	vendor B	the site whose cleaning schedule now overlaps the peak shift
Shepherdsville, Kentucky	Fathom Mobility	47	2025-03	41%	vendor C	manipulator-equipped robots are concentrated here
Reno, Nevada	Blue Mesa Components	29	2025-05	57%	vendor D	the smallest customer deployment
Newnan, Georgia	Ironwood Grocery Group	66	2025-07	33%	vendor B	narrow-aisle geometry, and the source of the August 2025 mutual-yield stall
Whitestown, Indiana	Juniper Bay Apparel	44	2025-09	45%	vendor C	the site with the standing map publication workaround
Bethlehem, Pennsylvania	Kestrel Medical Supply	37	2025-11	51%	vendor A	sparse survey features, which is why relocalisation is slow here
Chino, California	Larkspur Home	33	2026-01	54%	vendor D	the go-live delayed by the forked provisioning template
Edwardsville, Illinois	Crestline Fulfillment	24	2026-03	63%	vendor B	the newest Crestline site, still ramping
Lockbourne, Ohio	Granite Peak Distribution	19	2026-04	68%	vendor A	commissioned after the template baseline check landed
Mesquite, Texas	Evergreen Warehousing	16	2026-06	71%	vendor C	in acceptance testing at the review date

7. Security findings bearing on availability

SEC-2026-19 against Localization Engine is open, with a closure date of committed 2026-10-15. For Medium-severity finding SEC-2026-19 the record shows that cache read from a removed module, which is moderate evidence and is graded separately from severity. The residual position on SEC-2026-19 is medium; disk encryption is planned for the next compute revision, stated after mitigation rather than assuming full success. What Localization Engine finding SEC-2026-19 enables is this: a stolen compute module discloses a customer's warehouse layout. SEC-2026-19 against Localization Engine was reviewed against the position twelve months earlier, and the direction of travel is favourable.

Moderate, because the demonstration was performed in a laboratory tenant rather than in production evidence supports SEC-2026-05, namely that credential reuse confirmed from the secret store inventory. On 2026-04-16 the tester recorded Alarm Router finding SEC-2026-05: alarm router delivery provider credential shared across regions. Alarm Router finding SEC-2026-05 matters because one leaked credential affects paging for every region. Finding SEC-2026-05 arrives as a Medium finding with a 5.2 score, and the reviewer did not adjust either. assessment finding SEC-2026-05 is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

SEC-2025-11 against Remote Operations Console leaves residual risk that the reviewer assesses as low. SEC-2025-11 is rated Low with a numeric band of 3.1 on the tester's scoring scale. What assessment finding SEC-2025-11 enables is this: a cross-site request could reuse an active operator session. SEC-2025-11 against Remote Operations Console stands remediated; 2025-10-30 is what the record gives for closure. finding SEC-2025-11 was cross-checked against the engineering record supplied for TDR-2025-09, and the two agree.

The reason SEC-2026-04 was not downgraded is that customer commercial data is retained in operational logs. For SEC-2026-04 the reviewer notes that the item is open past its committed date and no revised date has been agreed. SEC-2026-04 was graded Medium, band 5.6, by Meridian Assurance engagement MA-2026-1. Medium-severity finding SEC-2026-04 is supported by the fact that payloads recovered from a staging log store; evidence strength is moderate, because the demonstration was performed in a laboratory tenant rather than in production. SEC-2026-04 was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

Reading Low-severity finding SEC-2026-12, the defect is that edge gateway agent retained mutual authentication keys after decommissioning; the discovery date is 2026-04-20. On finding SEC-2026-12 the conclusion is that the severity is right and the ownership is unclear, since two teams each believe the other holds it. What remains after finding SEC-2026-12 is remediated is medium; a wipe procedure exists but was not applied retroactively. The tester scored Edge Gateway Agent finding SEC-2026-12 at 3.5 and labelled it Low. SEC-2026-12 against Edge Gateway Agent was excluded from an earlier draft of TDR-2025-09 and reinstated after the reviewer obtained the supporting record.

What SEC-2026-20 against Payload Scale Interface enables is this: a calibration change cannot be reconstructed during an investigation. Moderate, because the demonstration was performed in a laboratory tenant rather than in production evidence supports SEC-2026-20 against Payload Scale Interface, namely that record edited on a laboratory robot. As at the review date Low-severity finding SEC-2026-20 was accepted and the closure entry reads not planned. The finding at Payload Scale Interface finding SEC-2026-20, dated 2026-04-24, is that payload scale calibration record can be edited without an audit entry. Payload Scale Interface finding SEC-2026-20 was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

SEC-2025-10 against Charge Scheduler produces this judgement: the item is open past its committed date and no revised date has been agreed. Residual exposure on Charge Scheduler finding SEC-2025-10 reads low. SEC-2025-10 against Charge Scheduler records that charge scheduler exposed an unauthenticated internal health endpoint with site names, discovered on 2025-09-24. The practical consequence in finding SEC-2025-10 is that site naming and fleet size leak to anyone inside the cluster network. Charge Scheduler finding SEC-2025-10 carries a follow-up request that was satisfied after the draft of TDR-2025-09 was circulated.

Robot shadow store cache lacked authorization on the internal read path is the substance of High-severity finding SEC-2026-02, raised on 2026-04-15. SEC-2026-02 is remediated, with a closure date of 2026-05-29. Finding SEC-2026-02 is assessed as follows: the exposure is internal rather than customer-facing, which lowers the severity without removing the finding. High is the rating SEC-2026-02 against Robot Shadow Store carries, at 7.6 on the assessor's own scale. SEC-2026-02 rests on a single source, and the reviewer has said so rather than implying corroboration.

Finding SEC-2026-06 sits against Simulation Harness and records that simulation harness cluster used a long-lived cloud role with broad read access. Medium-severity finding SEC-2026-06 is supported by the fact that role policy read from the cloud configuration export; evidence strength is strong. What SEC-2026-06 enables is this: a compromised scenario could read production artefact storage. SEC-2026-06 arrives as a Medium finding with a 6.1 score, and the reviewer did not adjust either. SEC-2026-06 was cross-checked against the engineering record supplied for TDR-2025-09, and the two agree.

At 3.9 the score on SEC-2026-09 against Fleet Health Monitor places it in the Low group. Fleet health monitor query interface allowed unbounded result sets is the substance of finding SEC-2026-09, raised on 2026-04-18. What finding SEC-2026-09 enables is this: a single query can exhaust the service's memory budget. SEC-2026-09 against Fleet Health Monitor leaves residual risk that the reviewer assesses as low. SEC-2026-09 was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

The reviewer's position on assessment finding SEC-2025-01 is that the remediation is adequate and the time to remediate is not. Substantiating High-severity finding SEC-2025-01 is the fact that reproduced by the tester and confirmed in the token audit, graded moderate. 2025-09-16 is the discovery date of High-severity finding SEC-2025-01, which finds that service tokens are issued with a twelve-hour lifetime. SEC-2025-01 against Identity and Authorization translates into the following exposure: a captured token remains usable long after the session that produced it. Identity and Authorization finding SEC-2025-01 would move from a concern to a verified defect if one further measurement were supplied.

The finding at finding SEC-2026-10, dated 2026-04-19, is that billing usage ledger export written to a bucket without object lock. Impact of Billing Usage Meter finding SEC-2026-10 is not theoretical; an export could be altered after the fact. The evidence for finding SEC-2026-10 is moderate: bucket configuration read from the cloud export. Billing Usage Meter finding SEC-2026-10 is remediated, with a closure date of 2026-06-05. finding SEC-2026-10 is treated as material to the investment case rather than as a routine engineering item.

8. Reliability debt and what it would cost to clear

The dependency structure places programme item TD-24 behind a customer maintenance window that is not yet agreed. The consequence of leaving TD-24 on Costmap Fusion undone is that it three modified files under a file-level copyleft licence appear unpublished. Costmap Fusion item TD-24 is the item to publish the modified files carrying file-level copyleft obligations, classified under IP hygiene. IP hygiene item TD-24 is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

Register item TD-17 appears in the register and is unfunded and dependent on capital rather than engineering effort. Sequencing of TD-17 on OTA Updater matters: it is ahead of two other items that depend on it. The register shows programme item TD-17 as in design, due 2026-11-30. OTA Updater item TD-17 is treated as material to the investment case rather than as a routine engineering item.

The reviewer's comment on support item TD-03 is that management ranked this behind the separation monitoring work. Whether programme item TD-03 happens depends on funding, and it is partly funded, with the second phase unbudgeted. TD-03 is sized at 1 engineer-quarter, or $128,000 at the company's loaded rate. register item TD-03 was inspected in the repository rather than described from the ticket, which is why the account is specific.

About tooling item TD-07 it should be said that the check is written but not enabled as at the review date. Effort on TD-07 on Simulation Harness is estimated at 0.5 engineer-quarters, costed at $68,000. Tooling item TD-07 is sequenced last in a three-item chain, so its date is the least reliable on the register. programme item TD-07 is the clearest instance in TDR-2025-09 of engineering practice being better than the documentation suggests.

The estimate for Edge Gateway Agent item TD-12 is 1 engineer-quarter and $144,000. Reading resilience item TD-12 against the funded plan, the site alert landed first; the buffer extension needs a storage change on the compute module. In progress is the status of Edge Gateway Agent item TD-12, targeted at 2026-10-31. programme item TD-12 was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

1 engineer-quarter is the estimate on Localization Engine item TD-16, which converts to $126,000. Exposure carried by TD-16 on Localization Engine is that it a stolen compute module discloses customer warehouse layouts. On

programme item TD-16 the reviewer records that depends on the next compute module revision. programme item TD-16 was tested rather than accepted, and the test is described in the working papers behind TDR-2025-09.

Left in place, the condition behind Localization Engine item TD-02 strong-copyleft exposure in shipped robot software plus an unsupported dependency. The dependency structure places register item TD-02 in parallel with the release-process changes. The reviewer's qualification on TD-02 is that the reviewer treats this as both an IP and a supportability item. TD-02 carries a follow-up request that was satisfied after the draft of TDR-2025-09 was circulated.

Sizing on TD-15 on Alarm Router gives 0.25 engineer-quarters and $34,000. Alarm Router item TD-15 is last in a three-item chain, so its date is the least reliable on the register, which is why the target date is later than the effort estimate implies. Whether Alarm Router item TD-15 happens depends on funding, and it is unfunded and dependent on capital rather than engineering effort. TD-15 is treated as material to the investment case rather than as a routine engineering item.

The dependency structure places TD-09 on Event Bus in parallel with the release-process changes. Under support, entry support item TD-09 is to upgrade the streaming platform before its support window closes. Status on TD-09 on Event Bus is committed; the target is 2027-06-30. Event Bus item TD-09 is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

Reading the programme plan, TD-13 on Localization Engine is behind a customer maintenance window that is not yet agreed. Register item TD-13 matters because it relocalisation exceeds the operator prompt timeout at sparse sites. 2026-12-31 is the target for performance item TD-13, whose status is in design. Localization Engine item TD-13 is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

9. Incidents with a safety dimension

In the reviewed period hazard HZ-12 in the register produced two near-miss reports from site staff. Where Occasional-exposure hazard HZ-12 bites is every robot variant currently in production. Reading register entry HZ-12 against the verification evidence, one site's dock layout leaves less clearance than the layout guide requires. Occasional-exposure hazard HZ-12 was cross-checked against the engineering record supplied for TDR-2025-09, and the two agree.

Status verified and residual risk Tolerable are what the register gives for Remote-exposure hazard HZ-19. The design measure recorded at hazard HZ-19 in the register is dock occupancy sensing plus a low approach speed inside the dock envelope. For Remote-exposure hazard HZ-19 the verifying activity was occupancy sensor bench test plus a witnessed dock approach. Serious injury hazard HZ-19 is the clearest instance in TDR-2025-09 of engineering practice being better than the documentation suggests.

The population exposed to hazard HZ-13 in the register is every site with a mezzanine or ramp. Reading register entry HZ-13 against the verification evidence, two sites use barriers below the guidance height and detection is then marginal. Obstacle detection bench test plus a site procedure review is what verified hazard HZ-13 in the register. hazard HZ-13 in the register would be a candidate for the first post-close engineering review under any reasonable plan.

Register entry HZ-02 was verified through site-specific commissioning test at each mapped blind corner. The three risk terms on Serious injury hazard HZ-02 are severity Serious injury, exposure Occasional and controllability Low. Serious injury hazard HZ-02 has produced no field event in the reviewed period, which is the strongest available test of the control. hazard HZ-02 in the register is treated as material to the investment case rather than as a routine engineering item.

Tolerable is the residual position on HZ-15, whose status is partially verified. Has produced no event, though the control has activated repeatedly is the incident history behind HZ-15. Remote-exposure hazard HZ-15 is addressed by egress routes marked as keep-out and a stop-and-clear behaviour on a fire signal, and the reviewer traced that control into the design record. hazard HZ-15 in the register has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

Mitigation for Equipment damage hazard HZ-16 is atomic update with rollback and a version consistency check before mission acceptance. Verification of hazard HZ-16 was by two rollback drills plus a fleet version audit. The parameter set for register entry HZ-16 reads Equipment damage, Remote, Low. hazard HZ-16 in the register was reviewed against the position twelve months earlier, and the direction of travel is favourable.

Evidence of verification at hazard HZ-09 in the register consists of ramp testing on a wet and a dry surface at two sites. The reviewer's note on hazard HZ-09 in the register is that the slip detector is prototyped and unvalidated as at the review date. Hazard HZ-09 in the register depends for its mitigation on ramp speed limits, a slip detector and a mechanical brake hold. HZ-09 was reviewed against the position twelve months earlier, and the direction of travel is favourable.

Severity at Equipment damage hazard HZ-22 is Equipment damage, with the exposure rated Occasional and controllability Medium. Register entry HZ-22 remains Tolerable for outages under the tested duration after the control is applied, and the register marks it partially verified. Protection at HZ-22 comes from local yield behaviour with a conservative speed limit and a timed stop. Occasional-exposure hazard HZ-22 was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

Fatality harm, Improbable exposure and Low controllability are the parameters recorded for Improbable-exposure hazard HZ-05. The hazard recorded at hazard HZ-05 in the register is that drive power is not removed when the emergency stop is actuated. Verification of hazard HZ-05 was by witnessed actuation test on every unit at build and at annual service. Improbable-exposure hazard HZ-05 was compared with the equivalent position at two other components, and it is neither the best nor the worst.

In the reviewed period Serious injury hazard HZ-18 has never activated in the field and is evidenced only on the bench. What the register does not say about hazard HZ-18, and the reviewer does, is that a design study to measure closing rate is running with a vendor assessment booked. The residual entry for register entry HZ-18 reads Not yet tolerable at the higher approach speed, open. HZ-18 was checked against the company's own register, which had not been updated since the change landed.

10. Unresolved matters and management's declinatures

Three of the open items in the present report would close on a single document that the company has agreed to produce. The open items in the present report are grouped by what would resolve them rather than by which team owns them, because several cross team boundaries. S. Alderton led the work behind report TDR-2025-09, combining document review, 19 interviews and hands-on inspection of running systems. This report follows the reviewer's usual practice of grading evidence strength separately from finding severity, so a plausible concern is not promoted to a verified defect. Calder Rowe Engineering Advisory has reported the position as at 2025-09-05 and has not attempted to predict where it will be at completion.

S. Alderton asked in every declined case who would revisit the decision and when, and the present report records the answer even where there was none. Where management has declined to act, this report preserves the decision and the stated rationale so a post-close plan can revisit it. Report TDR-2025-09 added two components to its sample after the first week, because the interview evidence pointed at them. Selection for this report favoured components that appear in the incident record, which biases the sample towards known trouble and is stated as such. The numbering in report TDR-2025-09 follows the order of the work rather than the order of importance.

Monitoring recommended by this report relies on metrics the company already emits, so no new instrumentation is required to start. Calder Rowe Engineering Advisory recommends that the 5 indicators be reported with their measurement basis attached, since two of them changed basis during the reviewed period. 4 items in TDR-2025-09 sit in a declined state, and the reviewer regards one of them as wrongly declined. 4 declined items are carried in this report, of which two lack any documented compensating control. The report at TDR-2025-09 does not restate the company's own reporting where the reviewer was able to verify the underlying data instead.

S. Alderton treats a backlog ticket as evidence of awareness and a merged change with a test as evidence of remediation, and the report at TDR-2025-09 keeps the two apart. The evidence convention in report TDR-2025-09 is that severity describes consequence and evidence strength describes confidence, and the two are never merged. The report at TDR-2025-09 records 4 items the engineering team declined to remediate, each with the reason given and the reviewer's view of it. Two positions in this report are unresolved between the reviewer and the engineering team, and both are left unresolved on the page. The present report was prepared on the footing that a reader may be an engineer or may not, and defines terms where the distinction matters.

11. Conclusions on reliability and post-close monitoring

Whatever weight a reader gives the individual findings, the present report concludes that the safety work is the strongest part of the engineering record and the delivery process is the weakest. The judgement this report reaches is that the safety work is the strongest part of the engineering record and the delivery process is the weakest. 4 items in this report are open, two of which the reviewer would treat as pre-closing information requests rather than post-close work. Where report TDR-2025-09 cannot conclude, it says what evidence would let it conclude. S. Alderton has recorded the date of each measurement, because several of them will be stale within a quarter.

Deal implications in TDR-2025-09 are modest in aggregate and specific in detail: 3 items for the closing checklist, 5 for the plan. For the transaction, report TDR-2025-09 identifies 3 matters the reviewer would raise before closing and 5 for the post-close engineering plan. The report at TDR-2025-09 is limited by log retention: the telemetry window is ninety days, so events before 2025-11-15 could not be examined directly. S. Alderton could not verify one class of claim in the report at TDR-2025-09, namely the behaviour of components deployed only at the two sites not visited. S. Alderton has avoided aggregating findings into a single score, because the aggregate would conceal the distribution.

TDR-2025-09 would have the monitoring pack include declined items with their compensating controls, so that acceptance decisions age visibly. The pack this report proposes is deliberately short, because a longer one would be produced and not read. Where the report at TDR-2025-09 says a defect is verified, the reviewer reproduced it or read a production trace of it; everything else is recorded as a concern. The report at TDR-2025-09 distinguishes three positions: verified defect, credible concern, and matter requiring further work. The third category is not empty. Calder Rowe Engineering Advisory was given read access without restriction on scope within the agreed subject matter.